

P A L E T T E

Pixel & Pages

PROJECT DETAILS

- **Name:** Kaviya S
- **Roll No:** E25AI118
- **Department:** B.Sc Computer Science With AI

ENTITY- RELATIONSHIP & SCHEMA DEFINITION REPORT

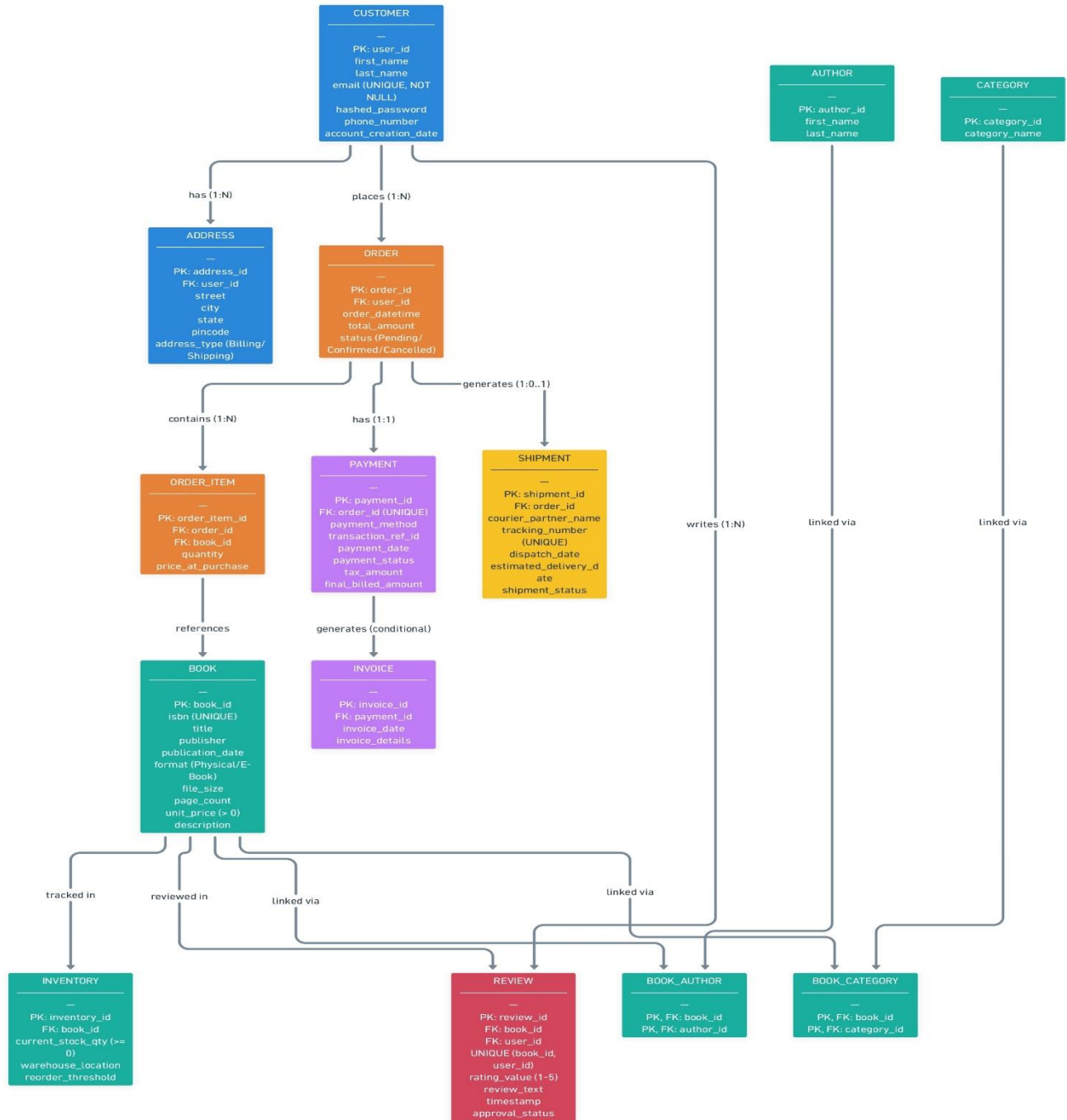
Project: E-Commerce Bookstore Database Management System

1. INTRODUCTION

This document represents the second phase of the E-Commerce Bookstore DBMS project. Following the initial Requirement Analysis, the goal here is to translate our business rules into a logical database structure.

A major focus during this design phase was ensuring the database complies with Third Normal Form (3NF). In an e-commerce setting, redundant data can lead to serious update anomalies (for example, if a user's address is stored on every single order, updating that address becomes a nightmare). To prevent this, data has been strictly separated into distinct entities. This report outlines that structure, mapping out the tables, their attributes, and how they connect to enforce referential integrity.

2.ENTITY-RELATIONSHIP DIAGRAM



3. DESIGN DECISIONS & NORMALISATION

Before diving into the data dictionary, a few specific design choices were made to translate the ER diagram into a functional SQL schema:

- **Historical Pricing:** The ORDER_ITEM table includes a price_at_purchase attribute. This is intentionally separated from the unit_price in the BOOK table. If a publisher raises a book's price in the catalog, we cannot let past invoices change.
- **Resolving Many-to-Many Relationships:** A single book can have multiple authors, and belong to multiple categories. Storing these as comma-separated values would violate First Normal Form (1NF). Therefore, junction tables (BOOK_AUTHOR and BOOK_CATEGORY) were created to handle these relationships properly.

4. ENTITY AND ATTRIBUTE DICTIONARY

The table below breaks down every entity, its primary key (PK), attributes, and the specific database constraints required.

Entity Name	Primary Key	Attributes	Implementation Notes & Constraints
CUSTOMER	user_id	first_name, last_name, email, hashed_password, phone_number, account_creation_date	email is UNIQUE and NOT NULL (used for login). Passwords are explicitly stored as hashes, never plain text.
ADDRESS	address_id	user_id (FK), street, city, state, pincode, address_type	address_type is constrained by a CHECK rule allowing only 'Billing' or 'Shipping'.
ORDER_MASTER	order_id	user_id (FK), order_datetime, total_amount, status	Tracks the overall cart checkout. status is limited to 'Pending', 'Confirmed', or 'Cancelled'.
ORDER_ITEM	order_item_id	order_id (FK), book_id (FK), quantity, price_at_purchase	quantity must be >= 1. Acts as the line-item breakdown for the Order_Master.

Entity Name	Primary Key	Attributes	Implementation Notes & Constraints
PAYMENT	payment_id	order_id (FK), payment_method, transaction_ref_id, payment_date, payment_status, tax_amount, final_billed_amount	Must maintain a strict 1-to-1 relationship with the order.
INVOICE	invoice_id	payment_id (FK), invoice_date, invoice_details	This table is only populated if the associated payment_status is marked as 'Success'.
SHIPMENT	shipment_id	order_id (FK), courier_partner_name, tracking_number, dispatch_date, estimated_delivery_date, shipment_status	tracking_number is UNIQUE. Digital e-book orders will skip this table entirely.
BOOK	book_id	isbn, title, publisher, publication_date, format, file_size, page_count, unit_price, description	isbn is UNIQUE. unit_price cannot be negative. format is restricted to 'Physical' or 'E-Book'.
INVENTORY	inventory_id	book_id (FK), current_stock_qty, warehouse_location, reorder_threshold	current_stock_qty must be ≥ 0 .
REVIEW	review_id	book_id (FK), user_id (FK), rating_value, review_text, timestamp, approval_status	A user can only review a book once (Composite UNIQUE constraint on book_id + user_id). rating_value is an integer between 1 and 5.
AUTHOR	author_id	first_name, last_name	Master list of authors to prevent spelling inconsistencies.
CATEGORY	category_id	category_name	Master list of genres (e.g., Fiction, Sci-Fi).
BOOK_AUTHOR	book_id, author_id	(Junction Table)	Composite Primary Key. Connects books to multiple authors.
BOOK_CATEGORY	book_id, category_id	(Junction Table)	Composite Primary Key. Connects books to multiple genres.

5. RELATIONSHIP MAPPING

This section defines how foreign keys connect the tables together, dictating the operational logic of the bookstore.

Source Entity	Relationship	Target Entity	Cardinality	How It Works in SQL
CUSTOMER	has	ADDRESS	1 : N	One user can save multiple delivery addresses. user_id is a Foreign Key in ADDRESS.
CUSTOMER	places	ORDER_MASTER	1 : N	One user can place many orders over time. user_id is a Foreign Key in ORDER_MASTER.
ORDER_MASTER	contains	ORDER_ITEM	1 : N	An order can have multiple books inside it. order_id is a Foreign Key in ORDER_ITEM.
BOOK	references	ORDER_ITEM	1 : N	A single book title can appear across thousands of different user orders.
ORDER_MASTER	has	PAYMENT	1 : 1	An order has exactly one payment record. order_id has a UNIQUE constraint in the PAYMENT table.
PAYMENT	generates	INVOICE	Conditional	Dependent on payment success. payment_id links the invoice back to the transaction.
ORDER_MASTER	generates	SHIPMENT	1 : 0..1	Physical orders get one shipment track. E-books get zero.
BOOK	tracked in	INVENTORY	1 : 1	Inventory levels are tied directly to the book_id.
CUSTOMER	writes	REVIEW	1 : N	A customer can write many reviews.

Source Entity	Relationship	Target Entity	Cardinality	How It Works in SQL
BOOK	reviewed in	REVIEW	1 : N	A book can receive many reviews from different customers.

6. CONCLUSION

By mapping out the exact tables, data constraints, and relationship cardinalities, we now have a complete logical blueprint of the e-commerce bookstore. Isolating the data into specialized tables (like splitting Authors and Categories away from the main Book table) ensures our backend is scalable and safe from data anomalies as the catalog grows.